Youssef Khalifa

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Available: 6-month engineering internship (end-of-studies thesis), February – July 2027

Profile

Final-year Software Engineering student at INSAT, building multi-agent AI systems and RAG pipelines on top of production backend engineering - LLM orchestration, retrieval services, containerised deployment, and CI/CD. Ships systems end to end, from architecture to running interface.

Experience

Backend Developer – Melyn May 2026 – Present

- Building and maintaining backend services and REST APIs in **.NET**, and contributing to **DevOps** work supporting the team's deployment and infrastructure.

Software Engineering Intern – WeeFizz Jun 2025 – Aug 2025

- Designed and implemented production REST APIs using **NestJS** and delivered a cross-platform mobile application with **Flutter**, covering end-to-end testing, system integration, and technical documentation.
- Developed and trained a CNN for automated fabric classification and an object detection model leveraging **OpenCV** and **TensorFlow**; engineered **web scraping** pipelines to automate dataset acquisition and annotation at scale.

Projects

QuantumBoard – Multi-Agent Quantum Circuit Design System 2026

- Designed a multi-agent system on a **blackboard architecture**: specialist agents (decomposition, synthesis, optimisation, verification) contribute incrementally to a shared circuit state under a control loop, rather than a fixed pipeline.
- Implemented circuit construction and execution with **Qiskit** against the **IBM Quantum** backend, including oracle construction and transpilation for real device topologies.

Stack: Python, Qiskit, IBM Quantum, FastAPI, Docker, React

Multi-Agent AI Travel Planner 2025

- Architected a multi-agent orchestration system in **n8n**: intake webhook → router → domain sub-workflows (flights, hotels, activities, local tips) → consolidated itinerary, powered by a **DeepSeek** LLM.
- Built per-domain **RAG** agents backed by a **FastAPI** + **FAISS** retrieval service, with **SerpAPI** and scheduled **Python** scraping pipelines (Playwright, BeautifulSoup) continuously harvesting and embedding live travel data.
- Restructured the orchestration graph for parallel per-city execution to eliminate workflow timeouts, and replaced blocking request/response with webhook acknowledgement plus **SSE** polling to a **Next.js** frontend.

Stack: n8n, Python, FastAPI, FAISS, DeepSeek, SerpAPI, Playwright, Next.js

DeepFake Detection System 2025

- Built an end-to-end video deepfake detection pipeline in **Python**: uniform frame sampling via **OpenCV**, face detection and spatial normalisation, feeding a fine-tuned **EfficientNet** classifier.
- Introduced a confidence-weighted voting mechanism across per-frame predictions to derive a video-level verdict, materially reducing false positives from isolated frame noise.
- Delivered a **web application** (Flask backend, responsive frontend) supporting video upload, inference, and visualisation of frame-level confidence scores and detection heatmaps.

Stack: Python, EfficientNet, TensorFlow/Keras, OpenCV, Flask

Technical Skills

- **Backend:** Node.js, NestJS, FastAPI, ASP.NET / .NET, Spring Boot, Flask
- **Frontend:** Next.js, React, Angular
- **Mobile & Game Dev:** Flutter, Unity
- **Databases:** MySQL, PostgreSQL, MSSQL – indexing strategy, triggers, stored procedures, CTEs, window functions
- **AI / ML:** Multi-agent orchestration, RAG pipelines, Vector DB (FAISS), TensorFlow/Keras, EfficientNet, Scikit-learn, OpenCV, Pandas, NumPy
- **Quantum:** Qiskit, IBM Quantum, oracle construction and circuit transpilation
- **Automation:** n8n, Python scraping (Playwright, BeautifulSoup), Firecrawl
- **DevOps & Cloud:** GitHub Actions, Docker, Kubernetes, Jenkins, AWS
- **Languages:** Python, TypeScript / JavaScript, C#, Java, C, C++

Education & Certifications

National Institute of Applied Science and Technology (INSAT) Sept 2022 – Present
Software Engineering Degree (in progress)

Baccalaureate – Mathematics, High School of Zaouia, Ksiba Thrayet **18.05/20**

DevOps using Jenkins & Kubernetes • AWS Cloud Practitioner (Beginner)